Emotion Recognition from EEG Signal using XGBoost Algorithm

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***Abstract*—Of late, emotion detection from brain signal has become a topic of research. Various machine learning algorithms have been applied to classify the emotion as a psychological measurement. However, good accuracy in measurements is yet to be achieved. This work proposes an approach to classify the different type of estimation in human being with improved accuracy. This work extracts several features from EEG brain signals and then optimize the set of features using the correlation matrix, information gain calculation, and recursive feature elimination method. The optimized feature set is then used in classification. For the classification technique, the Extreme Gradient Boosting (XGBoost) algorithm has been followed. The XGBoost algorithm works on the Gradient Boosting framework and follows linear model solver and tree learning concept. The results show that feature optimization followed by the XGBoost algorithm gives better classification accuracy. The proposed approach has been tested with DEAP data set. The proposed approach also has been compared with different classification algorithms, namely Naive Bayes, KNN, C4.5, Decision Tree, Random Forest algorithms. The results substantiate the efficacy in favor of the proposed approach.**

I. Introduction

A machine can do almost everything a human can do, even sometimes performs smarter. But, some works a human could still do better than the machine because of the psychological aspect of the human brain where emotion plays an important role. Till now, the machines were unable to process the emotion accurately like a human brain. So it's a challenging job to process the emotion through a machine or a robot. So the first step towards this development is the method of emotion recognition by a machine which is nowadays recognized as a part of affective computing [1]. Also, emotion influences human health conditions, for example, stress and anxiety can lead to serious health issues or depression, in the worst case towards death. Other application areas of emotions include the understanding of children psychology, education field, defense section, retailing, gaming, fatigue detection of a driver/pilot, etc. The list is pretty long. Many research works have been done recently on emotion recognition using EEG signals. Emotion recognition can also be done from the text [2], speech [3], facial expression [4] or gesture [2], [5]. Dan Nie et al. have used a minimum redundancy-maximum relevance method is used for extracting common critical features across subjects and Support Vector Machine (SVM) to classify EEG signal collected from subjects while they were watching movies [6]. Yuan-Pin Lin et al. have also used SVM to classify 4 emotion states when the subjects are listening to music and obtained an average classification accuracy of 82.29% $\pm$ 3.06% across 26 subjects based on 60 features derived from all EEG components with the feature temporal resolution of one second [7]. Zhang et al. proposed an emotion understanding system that classified user's states into two emotional states with the accuracy of 73% during image viewing using SVM classifier [8]. They extracted EEG features for binary emotion classification by combining the empirical mode decomposition and sample entropy methods. Apart from SVM various machine learning approaches has been used to recognize the emotional state of a person through EEG signal [9] [10].

Nevertheless, the works mentioned above got the best accuracy for all the emotions. But those works concentrated only on valence and arousal or just valence only. The work mentioned earlier either considered all of the features or applied one feature selection algorithm, which is time- consuming and not sufficient for such a huge number of features. For real-time application, the model for emotion recognition should be fast and accurate, and for that only the important features which contain relevant information are necessary. And also the existing works have been tested with only two or three classifiers which may not be sufficient to conclude.

Unlike others approaches, in this paper, a feature based classification model where a wide set of emotions based on 4 scales are classified considering the data generated by all 32 channels. Further, with the novel feature selection method followed by XGBoost classification algorithm, better accuracy can be achieved compared to recent works in all four scales. The proposed feature selection and classification method has been validated on the DEAP data set.

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II. Methodology

An overview of the proposed approach is shown in Fig. 1. Details of each step are discussed in the following.

A. Data Segmentation

EEG acquisition time a little longer than the accurate recognition time [11]. So for getting accurate results, EEG signals are usually separated into parts or segments by windows. But the size of the window is still a debatable topic. H.Candra et al. tested the DEAP data set with different window size, and they got the best accuracy $(65.33\% \pm 1.1\%)$ with 3s sample [12]. So in our work segmentation epochs of 3 secs (i.e., 384-time stamp) have been created. The length of each music video is 60 secs, and 32 participants are there. So, for each participant and each music video there are 20 epochs $(60/3)$. Therefore for all participants and all music video, there are $20 \times 32 \times 40$ (i.e., 20480) epochs.

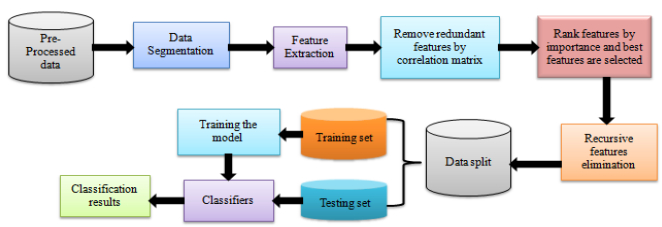

Fig. 1. Overview of the method

B. Feature Extraction

Feature extraction is a process where some features are extracted which can be used to classify the data and each feature contains some relevant information about the signal.

TABLE I
LIST OF EXTRACTED FEATURES

Type of Feature	Name of Features	
Time domain Features	Coefficient of Variation, Fractal dimension, Kurtosis, Second Differential Mean/Max, Skewness, First differential mean/max, Mean ABS XCORR EEG-EEG	
Hjorth Parameters	Activity, Mobility, Complexity	
Autoregressive Parameters	Arburg AR mean, min and max, Arburg P mean, min and max, Arburg K mean, min and max	
Frequency domain features	FFT-based features calculated from the EEG spectra	FFT DELTA, FFT THETA, FFT ALPHA, FFT BETA, FFT GAMMA, FFT WHOLE
	FFT-based Spectral Features	FFT DT RATIO DELTA/THETA, FFT DA RATIO DELTA/ALPHA, FFT TA RATIO THETA/ALPHA, FFT DTA RATIO (DELTA+THETA) /ALPHA, FFT SEF , FFT SP-ROLL OFF
Wavelet domain features	Mean, max, min, median wav value, Skewness wav value, Kurtosis wav value, Wav band Relative energy, Entropy spectral wav , Wav zero crossing , Wav coeff of variation , Wav total energy, 1st diff wav mean,max,2nd diff wav mean, max , 2nd diff mean, max	

For this work a wide range of features significant to emotion recognition have been reviewed. In this work 52 well defined features (Table I) which are known to discriminate EEG signals have been considered. This process also reduced the complexity of the work and improved the performance of the classifier.

C. Feature Selection

A total of 52 number of features may not be necessary to give accurate classification. Some of them may be redundant, so removing them will boost the performance of the classifier. For this reason a feature selection procedure has been followed:

1) Removing redundant features: Some features are highly correlated with each other. Those highly correlated features should be removed as removing redundant features will improve the performance of the classifier. Various functions are available to remove those correlated features. In this work, correlation matrix is formed from the features, and the features which are highly correlated are identified. Features with an absolute correlation of 0.90 or higher have been removed.

2) Rank features by importance: The importance of features can be estimated from data by building a model. Random forest has a built-in mechanism to report on variable importance. It calculates information gain for all the features to calculate importance and scale it in 0-1. Information gain tells us which features are more important. To measure information gain, we need to calculate entropy and average entropy of the features. The more the value of entropy, the more the information content. It's a common way to measure impurities so that we can remove the less useful features.

3) Recursive feature elimination method (RFE): This method is used to eliminate dependencies and collinearity in the model. This is an automatic feature selection method and also very popular [13], [14]. The algorithm is configured to explore all possible subsets of the features. An estimator is given that assigns weights of the features (e.g. coefficient of the linear model) and the aim of the RFE algorithm is to select the features by repeatedly constructing a model with a smaller and smaller set of features until the whole feature vector set is used up. To evaluate the model Random forest algorithm is used on each iteration. This step helps to improve the performance of the classifier.

D. Classification

The threshold for valence, arousal, dominance, and liking was kept 4.5, and thus it has been treated as a binary classification problem.Then we did 5 fold-cross validation [15] for 5 different classification model with boosting technique and selected best features and then calculated accuracy for each one of them. In this work, we used XGBoost method for classification. Nowadays XGBoost is a very popular algorithm among data scientists because of its first learning property through parallel and distributed computing. Apart from that, XGBoost is an unexplored algorithm in the field of emotion recognition. So here in this work we explored this algorithm to get better accuracy. XGBoost means Extreme Gradient Boosting(XGboost), and it is similar to gradient boosting framework but more efficient. It has both linear model solver and tree learning algorithms. So, what makes it fast can do parallel computation on a single machine.

III. Experimental Results

A. Data Description

The downsampled, pre-processed, segmented version of DEAP data set [5] in pickled Python/numpy formats has been used. In this data-set 32 subjects $(S_1 - S_{32})$ are shown 40 music video clips that are supposed to stimulate various types of emotions. The duration of each video is 60s. A preparation

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time of 3s was added to each music video during the pre-processing phase, and hence the total time of each video changed to 63s. The EEG signals are recorded during watching the video clip by 32 channels which were placed on the scalp of each subject. After watching the video, the subjects are asked to rate the video clip by an online assessment tool which contents questionnaire to understand the state of emotion at that time.

B. Result 1 (Feature Selection Method)

Data contain features that are highly correlated with each other. Features with an absolute correlation of 0.90 or higher were removed. As a result,9 features were removed for each scale of emotion due to high correlation with other feature. Then some features are removed again based on the information gain of the features. Then in next step, the RFE method repeatedly constructed a model and removed features with low importance. For example, we calculated accuracy on the different subset and got maximum accuracy with 7 attributes for valence. The same process was followed for other scales of emotion too. The summary of feature selection after applying three methods for each scale of emotions is shown in Table II.

To test the effectiveness of each method of feature selection,

TABLE II
SUMMARY OF FEATURE SELECTION AFTER EVERY METHOD FOR ALL EMOTIONS

	Initial Features	Correlation	IG	RFE
Liking	52	43	15	11
Arousal	52	43	27	16
Valence	52	43	7	7
Dominance	52	43	20	18

the accuracies of the classifier were tested after applying each method. Random forest classifier is used for this experiment. First, we will check with 52 number of features. That means no feature selection is applied and we will feed all the 52 features to the classifier. Then the classifier is applied to the reduced number of feature set after each step.

TABLE III
COMPARISON OF PERFORMANCE AFTER EACH STEP OF FEATURE SELECTION

No. of features selected	Classifier	Valence	Arousal	Dominance	Liking	Average Time
Total no of features=52	Random Forest	0.73214	0.7023	0.69624	0.74149	1min
After removing redundant features	Random Forest	0.73521	0.7326	0.69789	0.74267	0.72min
After removing redundant features and selecting features based on IG	Random Forest	0.73987	0.72759	0.69789	0.74365	0.61min
After removing redundant features based on IG and applying RFE method	Random Forest	0.7403	0.73561	0.74477	0.75878	0.5min

TABLE IV
COMPARISON OF OUR WORK WITH RECENT STUDIES

Reference	No of channels	Class	Channel selection	Classifier	Accuracy			
					Valence	Arousal	Dominance	Liking
I.Wichakam et al. [16]	32	Binary	No	SVM(LOTO)	62.3%	62.9%	67.3%	NA
	10	Binary	Yes	SVM(LOSO)	64.9%	65.0%	66.8%	NA
FK Ahmed et al. [17]	32	Binary	No	DWPT-GA-LS-SVM	48.30%	54.83%	NA	NA
				DWPT-LS-SVM	46.33%	49.22%	NA	NA
				PSD+DLN(PCA+CSA)	55.07%	52.56%	NA	NA
				PSD+DLN	49.52%	49.52%	NA	NA
Koelstra S, et al. [5]	10	Binary	Yes	kNN	57.6%	62%	55.4%	NA
Atkinson et al. [18]	14	Binary	Yes	SVM	73.14%	73.06%	NA	NA
A.Mart et al. [19]	18	Binary	Yes	ANN	72.87%	75.00%	NA	NA
Our proposed Method	**32**	**Binary**	**No**	**XGBoost**	**75.97%**	**74.20%**	**75.23%**	**76.42%**

From Table III, it can be proved that the accuracy is slightly improved as the number of features is decreased. Also, the time to get the output is noticeably reduced with the reduced number of features. Though there is no revolutionary change in accuracy of the classifier with a reduced number of features the model become fast and cut out the time of response with a reduced number of features.

TABLE V
COMPARISON OF PERFORMANCES FOR CLASSIFICATION

Classifier	Valence	Arousal	Dominance	Liking
Bayesian	0.7219	0.7128	0.7328	0.7422
KNN	0.7173	0.7167	0.7289	0.7525
C4.5	0.7249	0.7174	0.7336	0.7420
Decision Tree	0.7247	0.7200	0.7371	0.7456
Random Forest	0.7403	0.7356	0.7447	0.7587
XGBoost	**0.7597**	**0.7420**	**0.7523**	**0.7642**

C. Result-2 (Comparison of performances for classification)

We did 5 fold-cross validation by using 5 different classification model with boosting technique and selected the best features and then calculated accuracy for each one of them. We applied 6 different classification model like Bayesian, KNN, SVM, C4.5, Decision tree, XGBoost (Extreme Gradient Boosting), and Random forest for all 4 classes and the best accuracy is from XGBoost(Table V).

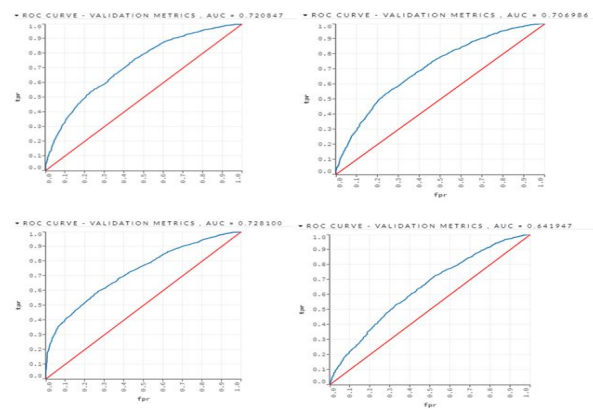

Fig. 2. ROC curves from proposed model for Liking, Arousal, Dominance and Valence.

The ROC curves from XGBoost model for valence, arousal, liking, and dominance are shown in Fig 2.

D. Result-3 (Related Work and Comparison with this work)

The existing methods tried to find the best method for emotion prediction by varying the number of channels or/and by using different methods of signal transformation or/and by applying different types of ML techniques. So we decided to compare our proposed method with them to make sure our method is giving better accuracy than the most promising classifier. In Table IV, we have listed all of the works mentioned earlier briefly, and the result indicates that with our feature selection method and classification technique the model can classify the emotion in four scales.

IV. CONCLUSION

The proposed method of feature selection improved the accuracy of the classification of emotions. This novel method of feature selection and XGBoost algorithm is giving 75.97%, 74.206%, 75.234% and 76.424% accuracy for valence, arousal, liking and dominance respectively which is higher than other standard classifiers accuracy. We hope that this study would be helpful in the future to explore and invent new methods for classifying human emotion, which can help to diagnose someone's mental state from their brain signal.

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